

Paper Reading Guide

miniGEM - 2025



Reading Scientific Papers

Adapted from S. Keshav, U Waterloo: The Three Pass Approach

The First Pass:

The first pass is a quick scan to get a bird's-eye view of the paper. You can also decide whether you need to do any more passes. This pass should take about five to ten minutes and consists of the following steps:

1. Carefully read the title, abstract, and introduction
2. Read the section and sub-section headings, but ignore everything else
3. Read the conclusions
4. Glance over the references, to see if you might recognize any (you probably won't for the first few papers in a category, don't worry!)

At the end of the first pass, you should be able to answer the five Cs:

- **Category:** What type of paper is this? A measurement paper? An analysis of an existing system? A description of a research prototype?
- **Context:** Which other papers is it related to? Which theoretical bases were used to analyze the problem?
- **Correctness:** Do the assumptions appear to be valid?
- **Contributions:** What are the paper's main contributions?
- **Clarity:** Is the paper well written?

Writers of papers expect most reviewers (and readers) to make only one pass over it. If a reader cannot understand the gist after one pass, the paper will be too obtuse or complex to be viable for greater audience; if a reader cannot understand the highlights of the paper after five minutes, the paper will likely never be read. Some papers are like this, and with those I wish you luck, unfortunately it doesn't get much easier to read the rough ones.

The Second Pass

The slogan of the second pass might be “read to understand.” Make notes or comments as you need, reading through with more detail specifically focusing on a few things:

1. Pay special attention to figures, diagrams, graphs and other illustrations. These items tell a lot about what the authors are trying to show you as the reader.
2. If a particular citation strikes your interest, mark it for further reading (this is a good way to learn more about the background of the paper).

After this pass, you should be able to summarize the main push of the paper, with supporting evidence, to someone else. Sometimes you won't understand a paper even after the second pass. This may be because the subject matter is new to you, with unfamiliar terminology and acronyms. The paper may even be poorly written with unsubstantiated assertions and a baseless claim or two (or it could just be that it's late at night and you're tired). Take a second to put the paper down before you decide whether a third pass is valuable to you and for your time.

The Third Pass

To fully understand a paper often requires a third pass. The key to the third pass is to attempt to re-implement the paper: that is, recreate the work with the author's assumptions. By comparing this with the actual paper, you can easily identify not only a paper's innovations, but also its hidden failings. Think about how you would present a particular idea. This comparison lends a sharp insight into the proof and presentation techniques in the paper and may help you write more. This pass might take a while, but it's shorter with experience. At the end of this pass, you should be able to pull the main ideas and findings from memory, and identify its strengths and weaknesses. Now, you have truly succeeded in breaking down a difficult academic paper. **Congratulations!!**